H.J. Baker & Bro., LLC

AP Invoice Processing Platform

Product Requirements Document

The product specification in one compact document: what the platform does, how each document type is handled, the rules it enforces, and how it is accepted.

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Prepared by Alferix

Document control

Document information

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1.0	31 Jul 2026	Alferix	First issue for client review
1.1	7 Aug 2026	Alferix	Focused specification: single end-to-end architecture diagram, per-document-type process steps, deduplicated against the SOW

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1 Introduction

H.J. Baker processes accounts payable for seven operating companies through a single mailbox and Sage X3. The current tool, Blue Creek, holds the document image and routes it for approval; the keying, coding and matching all happen by hand in Sage afterwards, and that manual effort is the largest cost in the process.

This platform replaces Blue Creek and automates the journey from arrival to payment. Every document is read and classified by AI, matched against Sage, routed for approval within the right limits, and posted back to Sage with its image attached, with people stepping in only where a genuine decision is required.

The platform is operated as a managed service: Alferix runs the accounts-payable pipeline from intake to posting, and H.J. Baker performs approval and final review. The requirements below describe the product regardless of who operates each step.

This document specifies the product. Delivery approach, timeline and commercials are set out in the Statement of Work; where the two differ, this PRD governs product functionality.

Table 1. *At a glance*

Replaces	Blue Creek (Vision360)
System of record	Sage X3, which stays the ledger; the platform never becomes a second set of books
Document types	PO, Non-PO, Prepayment, Credit, Rebill
Operating companies	Seven, across USD, CAD and GBP
Volume	About 3,000 to 3,300 invoices a month (from the Sage export sample)

2 Solution architecture

One route in, one classification decision, five processing paths, one system of record. A document arrives, is read and classified, is routed to the path for its type, is validated and matched against Sage, is approved within the right limits, and is posted to Sage, which returns the document number and, later, the payment.

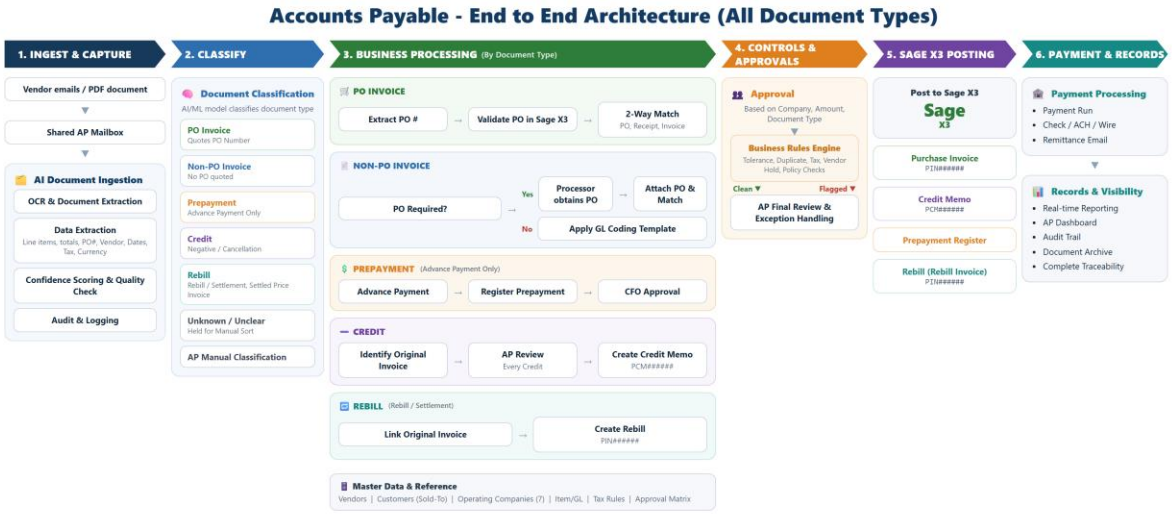


Figure 1. *Accounts payable end to end, across all five document types.*

Sage X3 stays the system of record throughout. Master data is read live at the moment it is needed, so the platform never works from a stale copy, and each document is written with its image attached in a single confirmed call.

Table 2. *The pipeline, stage by stage*

Stage	What happens
1. Ingest and capture	Ingest each attachment, split batch scans, refuse duplicates, and extract header and lines with per-field confidence
2. Classify	Route to one of five document types by rule; below the confidence threshold, hold for a person to classify
3. Business processing	Run the flow for the document type: match, code, register prepayments, or link settlement chains
4. Controls and approvals	Route by hierarchy and limit; apply the business-rule engine and AP final review where a rule fires
5. Sage X3 posting	Create the purchase invoice or credit memo with the image; capture the returned PIN or PCM
6. Payment and records	Capture payment status; keep a complete, searchable, auditable record

3 Core capabilities

Ten capabilities carry the platform.

Table 3. *Core capabilities*

Capability	What it delivers
Document intake	Ingests every attachment as its own record, splits batch scans, preserves the source email, and refuses duplicates on vendor, number and amount

Capability	What it delivers
AI extraction	Reads header and every line with a confidence score per field, with no line-count ceiling; reads the vendor live from Sage
AI classification	Routes to one of five types by a priority rule set; low confidence, or an unknown vendor, holds the document for a person
Validation	Confirms net plus taxes plus freight equals gross, checks the four key header fields against the image, and validates coding against live Sage lists
PO matching	Two- or three-way per the PO line, receipt-level on the ship date and BOL pair, with tolerance checks and coding pulled from the order
Approval workflow	Hierarchy and ad-hoc routing, per-company limits, automatic escalation, delegation and email approval; no self-approval
Sage integration	Live reads; purchase invoice and credit memo creation with attachment; document-number write-back; payment feedback; idempotent writes; daily reconciliation
Payment tracking	Captures payment number, date and method from Sage and closes the document; draws down prepayment balances
Search and audit	Searches header and line fields; every displayed count clicks through to its documents; a complete, append-only audit trail
Reporting	Volume, automation rate, cycle time, exceptions, approvals, prepayment positions and chain integrity, all clickable to source and exportable

4 Intake, extraction and classification

One route in, one classification decision, five destinations. Everything that follows in sections 8 to 12 assumes this has already happened.

4.1 Intake requirements

Table 4. *Intake, step by step*

Ref	Step	Control
FR-1	Poll the AP mailbox and ingest every attachment as its own document, preserving the source email for later reading.	Nothing is ingested twice; the email remains viewable from the document.
FR-2	Accept PDF documents only, single or multi-page, including scanned batches of several invoices in one PDF.	Batch scans are separated; each separated document is auditable back to the original file and page range.
FR-3	Refuse duplicates on the combination of vendor, invoice number and amount.	Catches genuine resends without blocking legitimate re-invoicing at a different amount. The refusal names the document already holding the number.
FR-4	Extract header fields and every line, with a confidence score per field.	No line-count ceiling. Fields below a per-field threshold are marked for the processor rather than silently accepted.

Ref	Step	Control
FR-5	Read the vendor record live from Sage at extraction time.	Terms, currency, pay method and remit detail always reflect Sage.
FR-6	Show the image beside the extracted data with each field's confidence visible.	A processor can verify a value against the document without leaving the screen.

4.2 Classification rules (FR-7, FR-8)

Rules are evaluated in priority order and the first match wins. The table is configuration, not code, the AP administrator adds, reorders, disables and edits rules, and changes are audited.

Table 5. *Routing rules in priority order*

Priority	Condition	Routes to	Why this order
10	Vendor is marked as prepaid	Prepayments	The vendor relationship overrides document appearance. "We could mark those two vendors, anything from them is a prepayment."
20	Negative total, or cancellation wording	Credits	A negative total is unambiguous and must not be posted as an invoice.
32	Settlement (netback) vendor re-invoicing at a settled price	Rebills	Only these vendors produce rebills; the marking lives on the supplier profile.
40	A purchase order number is present and valid	PO invoices	The strongest positive signal available.
99	Anything else	Non-PO invoices	The safe default: a person will decide whether an order is needed.

The confidence threshold. A single configurable threshold (default 0.80) decides whether the platform routes on its own or asks. Below it the document waits in Manual Review with the reason stated, poor scan, unknown vendor, ambiguous document type, and a person classifies it in a screen that shows the image on one side and the five types on the other. Selecting a type only selects it; a confirm action commits and moves to the next document, so a mis-click costs nothing.

An unknown vendor applies a fixed penalty to confidence, because a document from a vendor Sage has never seen is exactly the case where a confident-looking guess is most dangerous.

Rollout posture. Automatic routing is earned, not assumed. The threshold starts high, and the business raises automation company by company as accuracy proves itself, which is how Symantha asked for it.

4.3 Manual review edge cases

Table 6. *Manual review, edge case register*

Condition	Required handling
The scan is poor or the vendor name is unreadable	Held for a person with the reason stated rather than guessed at. A faxed handwritten ticket and a degraded scan end up in the same place.
The document is a statement of account, not an invoice	Held for a person. A statement listing several open items is not something to post, and the ambiguity is surfaced rather than resolved silently.
One email carries several invoices	Split into separate documents at intake, each classified on its own merits, all traceable to the one email.
The document is not an AP document at all	A person marks it as out of scope with a reason; it leaves the working queues but stays searchable and auditable.

5 Document types

The five document types each have their own trigger, their own checks, the point at which they must stop for a person, and a different result in Sage.

Table 7. *The five document types at a glance*

Type	Recognised by	Becomes in Sage	Always stops for
PO invoice	A valid purchase order on the document	Purchase invoice (PIN)	Approval
Non-PO invoice	No purchase order present	Purchase invoice (PIN)	The PO decision, then approval
Prepayment	The vendor is marked as prepaid	A payment, then invoices drawing it down	Approval, usually CFO for the lump sum
Credit	Negative total or cancellation wording	Credit memo (PCM)	The original invoice reference, then AP review
Rebill	A settlement vendor re-invoicing at a settled price	Purchase invoice linked to the credit	The credit-memo reference, approval, AP if the period is closed

5.1 PO invoices

Goods or services ordered properly, where the order already exists in Sage. Match type is read from the PO line (two- or three-way), receipts are matched on the ship date and BOL pair, and matched lines are coded from the order. Awkward cases handled: the invoice arriving before the goods (two-way, marked as such), a monthly summary covering a thousand receipts, a load list past a thousand lines, and a duplicate refused on vendor, number and amount.

Process steps

Table 8. *PO invoice, required behaviour, step by step*

Step	What happens	Actor	Control enforced
1	Header and lines are extracted with confidence per field; the order number is read.	Platform	Fields below threshold are marked, not accepted silently.
2	The order is validated against Sage: it exists, it is open, the vendor agrees, the company agrees.	Platform	A mistyped or closed order stops here with the reason named (FR-9).
3	Match type is read per line from the order, Sage carries 2-Way or 3-Way on the line itself.	Platform	The platform does not decide the match type; the order does.
4	For three-way lines, open receipts are selected by the ship date and BOL pair, which is the unique receipt key.	Platform	Receipt-level matching (FR-10). A thousand receipts are reconciled as a set; nobody ticks boxes.
5	Quantity and price are compared to the order within configured tolerance.	Platform	Outside tolerance becomes an exception with the variance shown, not an automatic rejection to the vendor.
6	Lines are coded from the order: account, division, sales channel, site, department.	Platform	No manual keying for matched lines. This is the coding saving.
7	Totals are validated: net plus taxes plus freight must equal the stated gross.	Platform	Cannot be submitted while out of balance; the difference is shown (FR-13).
8	The four key header fields are confirmed against the image, vendor, invoice number, date, amount.	Processor	Defaults from the supplier record never override what the document says.
9	The document routes for approval by hierarchy and limit, or ad-hoc for vendors not on the hierarchy list.	Approver	Auto-matched still requires approval unless AP has exempted it. No self-approval.
10	AP rules are evaluated. If none fire, the document goes straight to Sage; if one fires, AP reviews first.	AP team	Rules are configuration (Section 6) and every release is audited.
11	The purchase invoice is created in Sage with the image attached; the returned PIN is written back.	Platform	Idempotent on platform ID, a retry cannot double-post.
12	Payment status is captured from Sage and the document closes.	Platform	Payment number and method recorded on the record.

Edge cases

Table 9. *PO invoices, edge case register*

ID	Condition	Required handling
PO-01	The invoice beats the goods, nothing receipted yet	Match against the order alone and mark the document clearly as a two-way match, so the difference is visible to whoever approves it.
PO-02	Quantity or price does not agree	Exception for a processor with the variance stated. Most are resolved internally; rejection to the vendor is a decision, not a default.
PO-03	One invoice covers many bills of lading	Legitimate for some vendors. Those vendors sit on an approved list so the one-BOL check does not fight them every month.
PO-04	A monthly summary covers about a thousand receipts	Open receipts for the period are selected as a set and reconciled against the invoice total.
PO-05	A load list runs past a thousand lines	Processed whole, across however many pages it takes. Line count is not a limit, this is the Blue Creek ceiling removed.
PO-06	The same invoice arrives twice	Refused before creation on vendor, number and amount together, naming the document that already holds it.
PO-07	Taxes and freight do not add up	Submission blocked and the difference shown. GST, PST and VAT are held as separate lines, not lumped together.
PO-08	Canadian and UK invoices	Currency and tax treatment follow the company that owns the invoice, so each entity's tax lines are correct.
PO-09	Payment terms on the document differ from the vendor default	The document wins, because that is what was agreed on this purchase, and the override is flagged so AP can see it.
PO-10	The order is already fully invoiced	Refused with the invoice that consumed it named, so the processor can tell a duplicate from a wrong order number.
PO-11	Sage is unreachable at write time	The document holds in a retry state with the failure visible; it never silently disappears and never double-posts on recovery.

5.2 Non-PO invoices

An invoice with no purchase order. A person decides: attach an order that now exists, wait in a visible awaiting-order state, or code the lines from a standing template for a supplier that will never have an order (rent, equipment, services). The platform never creates the order; a mistyped or already-used order number is refused with the holder named.

Process steps

Table 10. *Non-PO invoice, required behaviour, step by step*

Step	What happens	Actor	Control enforced
1	Header and lines are extracted; no order number is found or the one present fails validation.	Platform	An invalid order number is reported as invalid, not ignored.
2	The document reaches the PO decision, a gate, not a guess. The vendor's last order in Sage is shown for reference.	Processor	The platform never creates a purchase order. Reference information only (Section 10, D-04).
3a	An order exists or is raised: its number is attached and validated against the numbering convention and against other invoices.	Processor	Wrong shape refused; already-used order refused with the holder named.
3b	This supplier will never have an order: lines are coded from a standing template built by the AP team.	Processor	Templates are configuration; a processor picks one and enters amounts only.
4	Where an order was attached, matching runs exactly as for a PO invoice.	Platform	Nothing is re-keyed; processing resumes where it stopped.
5	Totals are validated and the four key header fields are confirmed against the image.	Processor	Same controls as PO invoices.
6	Approval routes by hierarchy and limit, or ad-hoc for vendors not on the hierarchy list.	Approver	No self-approval; escalation is automatic above limit.
7	AP rules are evaluated, then the purchase invoice is created in Sage with the image and the PIN written back.	AP team, Platform	Idempotent write; AP release audited.

Edge cases

Table 11. *Non-PO invoices, edge case register*

ID	Condition	Required handling
NPO-01	An order is needed but does not exist yet	The invoice waits in a visible awaiting-order state with the vendor's last order shown, rather than disappearing into a pile on somebody's desk.
NPO-02	The vendor will never have an order	Office rent, equipment, professional services: lines are coded from a standing template and the invoice goes straight to approval.
NPO-03	The order number is mistyped	Checked against the numbering convention and refused before it reaches Sage.

ID	Condition	Required handling
NPO-04	The order is already on another invoice	Refused, and the invoice holding that order is named, so the processor can see at once whether this is a duplicate or a wrong number.
NPO-05	The ordering site raises the order days later	Attach it then. Processing resumes exactly where it stopped and nothing is re-keyed.
NPO-06	The scan is poor and the vendor is unreadable	Held for a person at classification rather than guessed at, with the reason stated.
NPO-07	The document is a statement of account	Held for a person. A statement of several open items is not something to post.
NPO-08	A template does not fit the invoice	The processor codes lines directly, and the gap is reported so the AP team can extend the template library.

5.3 Prepayments

Two vendors are paid before delivery, and their shipments draw that balance down. The lump sum registers a position and is approved, usually at CFO level; shipment true-ups draw it down and recalculate the consumed balance. Size alone never classifies a prepayment, treasury is warned as the balance runs low, and an overage is flagged as an unsecured payable.

Process steps

Table 12. *Prepayments, required behaviour, step by step*

Step	What happens	Actor	Control enforced
1	The vendor's prepaid marking routes the document here regardless of how the document looks.	Platform	The marking lives on the supplier profile and is AP-administered.
2	The document is classified as the lump sum or a shipment true-up, from its own wording and period, not from its size.	Platform	Size alone never decides, ordinary true-ups from these vendors run into eight figures.
3	Lump sum: a prepayment position is registered for the period, with the amount and the vendor.	Platform	A second lump sum for the same vendor and period tops up rather than duplicating.
4	Lump sum: approval routes at the level the amount demands, typically CFO.	Treasury, CFO	Limits apply exactly as elsewhere; nothing about scale bypasses the hierarchy.
5	Lump sum: it is paid and the payment is recorded against the position. Terms on these vendors can be as short as two working days.	Platform	Short terms are surfaced on arrival, not discovered late.

Step	What happens	Actor	Control enforced
6	Shipment: matched like an ordinary invoice, then drawn down against the open position for that vendor and period.	Platform	Consumed percentage recalculated on every drawdown.
7	Balance thresholds are evaluated: approaching the limit warns treasury; exceeding it flags for review with the overage stated.	Platform, treasury	The excess is an unsecured payable against a vendor already paid in advance, and is treated as such.
8	Approval, AP rules, Sage creation and payment feedback run as for any invoice.	Approver, Platform	Standard controls.

Edge cases

Table 13. *Prepayments, edge case register*

ID	Condition	Required handling
PRE-01	A second lump sum arrives for the same vendor	It tops up the existing position rather than opening a duplicate, and consumption is recalculated against the new total.
PRE-02	Shipments approach the prepaid amount	Treasury is warned while there is still time to act, not after the balance has gone.
PRE-03	Shipments exceed the prepaid amount	Flagged for review with the overage stated, because the excess is effectively unsecured credit to a vendor already paid.
PRE-04	A very large invoice arrives from a prepaid vendor	Size alone does not make it a prepayment. The document must say so; otherwise it is an ordinary true-up.
PRE-05	Payment falls due in two working days	Surfaced on arrival. A prepayment sitting unnoticed in a queue has a real cost with these vendors.
PRE-06	The document does not name its period	The period is inferred from the invoice date so the prepayment lands against the right month rather than being left unassigned.
PRE-07	A shipment arrives with no open prepayment	Treated as an ordinary invoice, matched, approved, posted, rather than failing because a balance was expected.
PRE-08	A quarterly true-up brings roughly three thousand invoices at once	Processed as a set within performance targets; the queue does not become unusable during a true-up window.

5.4 Credits

A document that reverses an invoice already booked, arriving with a negative total or cancellation wording. It must reference the original purchase invoice it reverses (Sage builds the credit memo from that invoice's lines), and every credit passes AP review. If a replacement invoice is expected, the credit parks visibly and clears when the rebill arrives.

Process steps

Table 14. *Credits, required behaviour, step by step*

Step	What happens	Actor	Control enforced
1	A negative total or cancellation wording routes the document to credits ahead of every other rule.	Platform	A negative document can never be posted as an invoice.
2	The original purchase invoice number is identified from the document.	Platform	Validated against the Sage numbering convention before anything is created.
3	If it is missing or malformed, processing stops and asks. The platform does not infer which invoice is being reversed.	Processor	Guessing would credit the wrong invoice, and that error compounds through the settlement chain.
4	The credit passes AP review. Every credit, including standalone goodwill adjustments.	AP team	Not rule-driven and not skippable (FR-16).
5	The credit memo is created in Sage, which pulls the lines from the referenced original invoice.	Platform	The returned PCM is written back onto the platform record.
6	If a replacement invoice is expected, the credit parks visibly in the queue AP uses for exactly this.	AP team	It clears automatically when the rebill arrives; nobody has to remember it.

Edge cases

Table 15. *Credits, edge case register*

ID	Condition	Required handling
CR-01	The credit does not say what it reverses	Processing stops and asks. Guessing would credit the wrong invoice and the error compounds down the chain.
CR-02	The referenced number is malformed	Checked against the Sage convention before anything is created, so a typo does not become a posted document.

ID	Condition	Required handling
CR-03	One credit covers several vendors	Credited invoices are identified individually and totals split per vendor, rather than one lump against whichever vendor is named first.
CR-04	The replacement invoice has not arrived	The credit parks visibly and clears automatically when the replacement comes in.
CR-05	The credit is genuinely standalone	A goodwill adjustment with no replacement still requires an AP release.
CR-06	A cancellation arrives from a settlement vendor	Treated as a credit first, because the negative total is unambiguous; its re-invoice is picked up separately as a rebill.
CR-07	The original invoice is in a closed period	Routed to AP with the closed-period condition stated. The platform does not decide period questions.

5.5 Rebills

Settlement vendors invoice provisionally, then re-invoice at the settled price. The rebill must name the purchase credit memo it settles, which links original invoice, credit and rebill into one provable chain (the gap the business named directly). A settlement reaching into a closed period routes to AP with that condition stated; a chain that is not yet complete states exactly which link is missing.

Process steps

Table 16. *Rebills, required behaviour, step by step*

Step	What happens	Actor	Control enforced
1	The vendor's settlement marking, plus a re-invoice at a settled price, routes the document here.	Platform	The marking lives on the supplier profile. A good share of these vendors are intercompany.
2	The purchase credit memo this rebill settles is captured in a dedicated field and validated.	Processor	Gate. Format enforced as PCM plus six digits; no reference means no chain (FR-17).
3	Original invoice, credit memo and rebill are linked as one chain, at the settled price.	Platform	All three Sage numbers on one record.
4	Matching runs against the purchase order as for any PO invoice.	Platform	Settled price replaces provisional; the difference is visible.
5	Approval routes by hierarchy and limit.	Approver	Standard controls.
6	If the settlement reaches into a closed accounting period, the document routes to AP with that condition stated.	AP team	The platform does not decide period questions; it presents them (FR-18).

Step	What happens	Actor	Control enforced
7	The purchase invoice is created in Sage and the returned PIN completes the chain.	Platform	Chain integrity is reportable: complete chains versus missing links.

Edge cases

Table 17. *Rebills, edge case register*

ID	Condition	Required handling
RB-01	Settlement reaches back into a closed period	The platform does not decide. It routes to AP with the closed-period condition stated, and waits.
RB-02	Settlement arrives in pieces across several months	The chain stays open and shows what is still outstanding, rather than appearing complete after the first piece lands.
RB-03	The rebill arrives before the credit	The chain shows the gap explicitly, so it is obvious a credit is expected rather than the rebill looking like an unrelated new invoice.
RB-04	The settled price is lower than the provisional	The rebill is smaller than the original and the net effect is visible on the chain, so the vendor position reads correctly at a glance.
RB-05	Somebody asks about it six months later	Original, credit and rebill are one linked chain with all three Sage numbers. That is the audit answer before the question is asked.
RB-06	The credit memo reference is mistyped	Refused against the Sage format before the chain is written, with the expected shape shown.
RB-07	A retro settlement reaches back three months across two companies	The chain records the company on each document so intercompany settlement remains reportable.

6 Cross-cutting logic and controls

Behaviour that applies to every document type, regardless of which flow it is in.

6.1 Matching (FR-7 to FR-11)

- Match type is taken from the purchase-order line, not decided by the platform. Sage carries 2-Way or 3-Way per line.
- Three-way matching is at **receipt level**, and the unique key for a receipt is the **ship date and BOL pair**, this is what makes a thousand-receipt monthly summary tractable.
- Quantity and price tolerances are configuration, set per company and optionally per vendor.
- A partial receipt matches partially, and the remaining quantity stays visible on the order rather than closing it.
- Unit-of-measure conversions on the order (pounds to long tons, for example) are applied from Sage's own conversion, never recalculated locally.

6.2 Coding and templates (FR-12, FR-14)

- Matched lines are coded from the purchase order. Unmatched lines are coded from a standing template, or by hand as a last resort.
- Templates are built and maintained by the AP team, not by developers. A template carries the full coding block and a description; a processor picks one, enters the amount, and submits.
- Every coding value is validated against the live Sage reference lists at submission, so a code retired in Sage cannot be posted from here.

6.3 AP final review (FR-15)

AP final review is a rule-driven control point. The rules are configuration and start from what the business already does today: amount above a threshold, a closed accounting period, every credit, a new vendor's first invoice, a variance released by a processor, and any vendor the AP team names.

> **Design note.** In session 002 the business stated "everything has to hit this final AP review before it comes into Sage." The platform is built rule-driven rather than universal, because universal review makes AP a bottleneck on every document and removes most of the benefit. The two are one configuration change apart.

6.4 Duplicate and integrity controls

- Duplicates are refused on vendor, invoice number and amount together (FR-3).
- A purchase order already consumed by another invoice cannot be attached again, and the holder is named.
- Net plus taxes plus freight must equal gross before submission (FR-13).
- Writes to Sage are idempotent on platform ID; a timeout retry cannot create a second document (NFR-6).
- Nothing is deleted. Cancellation and out-of-scope marking are states with reasons, and the record remains searchable.

6.5 Search and the click-through contract (FR-19 to FR-20)

Search covers header fields with exact, wildcard and range operators, and line-level fields where the business needs them, account, division, project, site.

Every count displayed anywhere in the platform, a dashboard tile, a queue badge, a sub-queue chip, carries the exact filter that produced it, and clicking it lands on those documents. This is a hard contract, not a convention: a count you cannot click through to is a defect. In the prototype it is enforced by 539 automated invariant checks on every displayed number.

6.6 Audit trail (FR-21, FR-22)

- Every state change, field edit, approval, rejection, release and Sage write is recorded with timestamp, actor and message.

- The document's own history is readable from the document: when it arrived, when it was classified and at what confidence, every processing step, who approved it, which Sage numbers were created and when payment was confirmed.
- Configuration changes, routing rules, thresholds, limits, templates, users, are audited with before and after values.
- Audit records are append-only and retained per the retention policy (NFR-9).

7 Business rules

Rules that govern routing, checks and posting. They are configuration the AP team owns, and every change is audited.

Table 18. *Business rules*

Ref	Rule
BR-01	Every document belongs to one of seven companies, determined from the sold-to name via a client-owned mapping; unknown names are held for a person, never guessed.
BR-02	Documents from a marked prepaid vendor route to Prepayments regardless of appearance.
BR-03	A negative total or cancellation wording routes to Credits ahead of every other rule; a negative document is never posted as an invoice.
BR-04	A marked settlement vendor re-invoicing at a settled price routes to Rebills.
BR-05	A valid purchase order routes to PO invoices; everything else defaults to Non-PO.
BR-06	Below a configurable confidence threshold, a document waits in Manual Review; an unknown vendor lowers confidence.
BR-07	Duplicates are refused on vendor, invoice number and amount together.
BR-08	Net plus taxes plus freight must equal gross before a document is submitted.
BR-09	The two- or three-way match type is read from the Sage purchase-order line; a receipt is identified uniquely by ship date and BOL.
BR-10	The platform never creates a purchase order; a person attaches one or codes from a template.
BR-11	A credit memo must reference the original purchase invoice it reverses; a rebill must name the credit memo it settles.
BR-12	Every document is approved; an auto-matched document still needs approval unless AP exempts it, and nobody approves their own submission.
BR-13	AP final review fires on rules: amount, closed period, every credit, a new vendor, a released variance, or a named vendor.
BR-14	Writes to Sage are idempotent on the platform document number; nothing is deleted, and cancellation is a state with a reason.

8 Non-functional requirements

Table 19. *Non-functional requirements*

Ref	Requirement	Target
NFR-1	Interactive response	Under 1.5 s at the 95th percentile at production volumes
NFR-2	Large document handling	A 1,000-line document opens, matches and submits whole; no line ceiling
NFR-3	Throughput	Sustained 3,500 documents a month, with true-up peaks, without queue degradation
NFR-4	Extraction turnaround	Median under 2 minutes from arrival to extracted and classified
NFR-5	Authentication	Single sign-on with MFA enforced by policy; no local passwords
NFR-6	Write integrity	Idempotent Sage writes; a failed write surfaces as a visible state, never a silent loss
NFR-7	Authorisation	Role and company scoped; a user sees only the companies they are entitled to
NFR-8	Data protection	Encrypted in transit and at rest; secrets in a managed vault; no financial data in logs
NFR-9	Retention	Documents and audit retained 7 years by default, configurable per company
NFR-10	Accessibility	WCAG 2.1 AA, verified by automated audit
NFR-11	Browser support	Current Edge, Chrome, Firefox and Safari; usable on tablet for approvals

9 Acceptance criteria

The platform is accepted when each area behaves as below on production-representative data.

Table 20. *Acceptance criteria*

Area	Accepted when
Intake	A batch scan splits into separate, traceable documents, and a duplicate is refused naming the document that already holds the number.
Extraction	A 1,000-line document is read whole, and fields below the confidence threshold are flagged rather than accepted silently.
Classification	Sample documents route to the correct queue, and a below-threshold document waits in Manual Review with the reason stated.
Validation	An out-of-balance document cannot be submitted, and a coding value not present in Sage is refused.
PO matching	A three-way invoice with receipts present matches and codes with no human edit, and a price variance appears as an exception.

Area	Accepted when
Approval	An over-limit amount escalates automatically, self-approval is blocked, and an email approval is recorded on the document.
Sage integration	A purchase invoice and a credit memo are created with the image and number written back within seconds, and reconciliation flags any divergence.
Credit and rebill	A credit cannot post without its original invoice reference, and a rebill chain shows all three Sage numbers or names the missing link.
Search and reporting	Every displayed count clicks through to the exact documents, and every view exports to spreadsheet.

10 Open decisions

Questions whose answers change the build. Each needs an owner and a date before the corresponding work starts.

Table 21. *Open decisions*

Ref	Decision	Owner
D-01	Confirm the Sage X3 API surface and field-level contract (our layouts are inferred from the export samples)	Net at Work with Alferix
D-02	Purchase-order numbering: is the sequence global or per company?	Net at Work
D-03	Approval hierarchy and per-company limits, as data	AP team
D-04	The template library: which suppliers, which coding	AP team
D-05	Sold-to name to company code mapping, including trading names	Symantha Schaeffer
D-06	Email approval: keep it, and for whom?	Symantha Schaeffer
D-07	Tolerance thresholds for quantity and price variance	AP team

Purchase-order creation is retired from scope: it is not automatable, so the platform asks for the order number and shows the vendor's last order for reference.

11 Glossary

Table 22. *Terms as used in this document*

Term	Meaning
Blue Creek	The incumbent tool (Vision360) this platform replaces
BOL	Bill of lading; with the ship date, it uniquely identifies a receipt

Term	Meaning
Netback / settlement vendor	A vendor who invoices provisionally then settles at a final price, producing a credit and a rebill
PIN / PCM	Sage purchase invoice / purchase credit memo
Prepayment true-up	A shipment invoice that draws down a prepayment
Rebill	A re-invoice at the settled price, following a credit
Straight-through	A document that reaches approval with no human edit
Two- / three-way match	Invoice against order, or invoice against order and receipt